(Please write your Exam Roll No.) *Exam Roll No. .....*

END TERM EXAMINATION

FIRST SEMESTER (B.TECH) JANUARY-2024

Paper Code: ES-119 ***Subject: Manufacturing Process***

Time: 3 Hours ***Maximum Marks :60***

Note: Attempt five questions in all including question no.1 which is compulsory. Select one question from each unit. Assume suitable missing data, if any.

Q1 **Short questions:-** **(2x10=20)**

(a) Explain various constituents of molding sand.
(b) Differentiate between hot working and cold working.
(c) Differentiate among Stiffness, ductility, and toughness with some examples.
(d) How is forging different from casting with respect to properties obtained in a product?
(e) Explain the term DCSP and DCRP in arc welding.
(f) What is spring back in sheet metal.
(g) Why flux is not required in electric resistance welding.
(h) What is edge preparation in Welding.
(i) Differentiate between thermosetting and thermoplastic polymer with some examples.
(j) What is rotational moulding of plastics? Name the parts made by rotational moulding.

UNIT-I

Q2 (a) What is die casting? Explain the difference between hot chamber and cold chamber die casting with neat sketch. **(5)**

(b) Discuss five major casting defects along with their remedies. **(5)**

Q3 (a) Discuss the five types of pattern allowances considered in casting. **(5)**

(b) Explain any five tools used in moulding process with diagrams. **(5)**

UNIT-II

Q4 (a) What is the difference between TIG and MIG welding. Explain with diagrams and their applications. **(6)**

(b) What do you mean by weldability? What are the pros and cons of welded joints over other joints. **(4)**

Q5 (a) Explain with diagram the working, advantages, limitations and applications of submerged arc welding. **(6)**

(b) Explain any four welding defects with its remedies. **(4)**

UNIT-III

Q6 (a) Sketch and explain the difference between hydrostatic extrusion and tube extrusion with their applications. **(5)**

(b) Explain any five tools used in sheet metal shop with diagrams. **(5)**

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Q7 (a) With the help of neat sketches explains the difference among the drop, press and machine forging. **(6)**

(b) Explain the difference between embossing and coining giving sketches. **(4)**

UNIT-IV

Q8 (a) Discuss the working principle of compression molding with its pros and cons. What are the various factors affecting compression molding. **(6)**

(b) Differentiate among blending, compacting and Sintering. **(4)**

Q9 (a) Explain conventional single stage plunger type injection molding with its neat sketch. **(5)**

(b) What are the different materials used for processing plastics. Explain any four with its functions. **(5)**

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